CHEMISTRY
P525/1
September, 2026

MBALE SECONDARY
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Sign:.....

DEPARTMENT OF CHEMISTRY
CHEMISTRY PAPER ONE

TIME: 2 HOURS 45 MINUTES

INSTRUCTIONS

- The paper consists of section A and B.
- Section B consists of two parts I and II
- Answer all items in section A and one item from each part of section B.
- Illustrate your answers with equations where applicable.

SECTION A

Attempt all items in this section

1. Heat changes during chemical processes are essential for managing thermal safety in extreme environments. An environmental engineering team is setting up an automatic weather station in a very cold zone. The station needs an emergency backup battery to power a radio transmitter so it can send weather data back to headquarters. To keep the radio running without losing power too quickly, the battery reaction of the chosen battery must release a minimum of 300.0 kJ/mol of electrical energy.
To find the best option, the engineering group builds two different battery cells in the laboratory.
Battery system 1:
$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$
Battery system 2:
$Zn(s) + 2Ag^{+}(aq) \rightarrow Zn^{2+}(aq) + 2Ag(s)$
Engineers noticed that due to the extreme low temperatures, the liquid inside the batteries had frozen and would not conduct electricity. To fix this, they installed a small heater inside the battery box to trigger the combustion of carbon in the presence of hydrogen so as to yield an intense amount of heat which melts the ice inside the battery to restore the liquid electrolyte back to the functional state.
$C(s) + 2H_2(g) \rightarrow CH_4(g)$
If a freeze warning occurs, the station's computer automatically feeds exactly 2.0 moles of carbon into the heater. For this safety system to work

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effectively and melt the ice, at least 140.0 kJ of total heat energy must be supplied by the heater.
The team has the following known data equations from laboratory records:

Equation	Enthalpy change of the reaction
$C(s) + O_2(g) \rightarrow CO_2(g)$	-393.5 kJ/mol
$H_2(g) + 1/2\ O_2(g) \rightarrow H_2O(l)$	-285.8 kJ/mol
$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + H_2O(l)$	-890.3 kJ/mol

Laboratory Data and Constants Pack

Half cell	Reduction potential
$Zn^{2+}(aq) + 2e^- \rightarrow Zn(s)$	-0.76 V
$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$	+0.34 V
$Ag^+(aq) + e \rightarrow Ag(s)$	+0.80 V

a) Analyse the data to determine the spontaneous nature of each battery system, select the best battery system for use, and comment on the effectiveness of the heater used.

b) Explain how the engineers can increase on the speed at which the ice is melted.

2. A pharmaceutical company supplying regional hospitals in Uganda manufactures an intravenous antibiotic that is highly effective only within a narrow pH range 4.8–5.3. Clinical trials showed that small changes in pH accelerated decomposition of the active ingredient, reducing its therapeutic value. To improve stability, chemists needed to develop formulations to use. However, during long-term storage some batches developed white precipitates that blocked intravenous tubing. Further investigations showed that calcium ions in the formulation reacted with carbonate ions introduced through poor storage conditions, forming insoluble calcium carbonate.
To identify the most suitable formulation, two of them were prepared using ethanoic acid (CH_3COOH) and sodium ethanoate (CH_3COONa). Each solution was prepared by dissolving the stated masses in distilled water and making the solution up to 250 cm^3.

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Table 1: Formulation preparation

Formulation	CH_3COOH (g)	CH_3COONa (g)
Q	4.50	2.05
R	1.50	6.15

Ka for $CH_3COOH = 1.8 \times 10^{-5}$

Table 2: Data obtained after storage.

Batch	$[Ca^{2+}]$ (mol dm^{-3})	$[CO_3^{2-}]$ (mol dm^{-3})
A	1.0×10^{-3}	2.0×10^{-6}
B	2.5×10^{-3}	3.0×10^{-6}
C	4.0×10^{-3}	2.0×10^{-5}

Ksp for $CaCO_3 = 4.8 \times 10^{-9}$

Tasks

a) Analyse the data provided and use it to recommend the most suitable formulation for preserving the stability of the intravenous antibiotic and the batch which doesn't endanger drug quality and patient safety.

c) Explain the importance of such formulations in real life, and the quality assurance measures that would minimise precipitation while maintaining drug effectiveness.

SECTION B

Attempt two items from this section one from each part

PART I

3. A company which manufactures cleaning agents use halogens such as chlorine because of its unique properties while a nearby factory uses carbonates of group II as its raw material. S.5 students visited the two, and realized that fluorine and chlorine behave differently when dissolved in water and alkalis while the carbonates dissolved in hydrochloric acid. The tour guide told them that the halogen halides are useful as acids depending on the bond energies. The information for bond energy was provided below.

Bond Energy Data

compound	HF	HCl	HBr	HI
Bond energy (kJ/mol)	562	431	366	299

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The students of S.5 were confused about the findings and have come to you as a S.6 student of chemistry;

Task

Help them to:

a) Explain the order of importance of the compounds mentioned as acids.

b) Analyze and explain the behaviours of the elements and compounds when dissolved as discovered by the students during the tour.

c) Assess the application of the halogen mentioned as a cleaning agent.

4. Many elements are widely used in everyday life because of their different physical and chemical properties.

During a community industrial and environmental awareness exhibition organized by a nearby university, learners from a certain school visited different demonstration stalls.

- At one of the stalls, workers demonstrated how Aluminium containers and phosphorous are cleaned using sodium hydroxide solution during recycling and maintenance processes.
- At another stall, emergency response officers demonstrated how magnesium flares are burnt to produce bright light during rescue operations.
- At the public health stall, chlorine gas was dissolved in water to demonstrate the purification of drinking water and lead is also dissolved in water to demonstrate how lead pipes dissolve to contaminate water.
- At another stall, the elements were heated strongly to obtain their melting points.

The researchers provided the learners with the following information about selected elements.

Property	Magnesium	Aluminium	Iron	Phosphorus	Chlorine
Melting point (°C)	649	661	1538	44	-101

After the exhibition, the learners were asked to interpret the observations made during the study tour, and they approached you for help.

Task

(a) Analyse the trend in melting points of the elements provided.

(b) Explain the chemical processes that may occur at the different stalls.

(c) Assess the usefulness of any of the two elements in society and measures that can be adopted to minimize problems associated with their use.

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PART II

5. An agrochemical manufacturing plant located at Luzira Industrial Area synthesises a fumigant used in post-harvest grain preservation. The active ingredient in this formulation is 1,2-dichloroethane and is dissolved in high-purity propanone as the industrial solvent to completely dissolve the pesticide.

During a routine supply audit, a new quality assurance intern raised two management questions to the technical team.

(i) The intern noted that instead of disposing of the raw expired batches of 1,2-dichloroethane pesticide, the factory could study how its chemically transformed in to ethyne.

(ii) The intern discovered an unlabelled container in the solvent storage bay. To verify whether it was the required propanone solvent, they suggested sampling the liquid and testing how it reacts with Brady's reagent (2,4-dinitrophenylhydrazine).

Furthermore, due to shortages in supply, the research division is investigating green chemistry pathways to make both components locally from locally available raw materials. They want to map out how the pesticide (1,2-dichloroethane) can be derived from chloroethane, and the solvent (propanone) can be efficiently produced starting from ethanol.

Understanding of the boiling points of the components and their raw materials as well as evaluating the use of propanone as the industrial solvent is important.

Task:
Make a report with guidance on the issues raised.

6. A certain company that manufactures alcohols and polymer-based packaging materials has recently expanded its production.
The different production units use hydrocarbons obtained from petroleum as raw materials to manufacture the alcohols and polymers. Due to increased production, there is scarcity of raw materials and they have been advised that Methyl-2-methylpropenoate a raw material for perspex and ethene a raw material for a plastic polymer polythene can each be made using propanone as the starting material. In order to be able to control likely explosions, the company is seeking to understand how the reactions for formation of polythene from its raw material and 1-bromopropan-2-ol from propene take place.
The company has been conditioned to effectively manage disposal of waste materials from the plastic polymer which is said to be non-

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biodegradable and this has prompted them to seek to understand why it behaves this way and evaluate its use. The company lacks qualified personnel to help them do all the above and has advertised for vacancies for those who can help. Your teacher who read the advert has provided you with this information.

Task

As a chemistry student, make a detailed write up you would use to try to help this company.

END

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